

vasic-digital tmux — Open Issues Tracker

Canonical source of truth for everything currently unfinished, partially validated, or at risk of violating the anti-bluff covenant (Constitution.md §1 + §11.4.1 through §11.4.6).

Every PASS in this codebase **MUST** carry positive evidence captured live that the feature works for the end user. Metadata-only PASS, configuration-only PASS, "absence-of-error" PASS, and grep-based PASS without runtime evidence are all critical defects regardless of how green the summary line looks. **Tests AND HelixQA Challenges are bound equally.**

Forensic anchor — direct user mandate (verbatim, 2026-04-28 + 2026-05-07 + 2026-05-08, repeatedly reasserted from upstream vasic-digital projects):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!"

§11.4.6 forensic anchor (verbatim, 2026-05-08):

"'LIKELY' is guessing, we **MUST NOT** have guessing, since it can be or may not be! No bluffing and uncertainty is allowed at any cost! We **MUST** always know exactly precisely what is happening exactly, in any context, under any conditions, everywhere!"

Compiled: 2026-05-08 (Phase B per-session-containerization cycle). **Author:** Engineering coordinator **Working pool source:** the items below feed the active task list directly. Each item carries a current state, the captured-evidence requirement, and a fix-direction proposal so future-self can resume cold.

Migration policy (mirrors upstream vasic-digital projects): resolved items are moved to **Fixed.md** in the same commit that closes them. **Issues.md holds OPEN / PARTIAL /**

BLOCKED / RUNNING / INVESTIGATED only; once an item is closed and verified end-to-end, it migrates to `Fixed.md` and disappears from here. Never delete items outright — history matters for cold-start handover.

Document conventions

Code	Meaning
OPEN	Unfinished work; needs implementation + anti-bluff coverage
PARTIAL	Implementation exists but coverage gaps remain (positive evidence missing or environmental SKIP unresolved)
BLOCKED	Cannot progress without external dependency (host capability, third-party tool, distro support, etc.)
RUNNING	In-flight in this session (background process, ongoing test cycle)
INVESTIGATED	Forensic investigation produced findings; closure pending decision on fix scope

Status reclassification rules (§11.4.6 enforced):

- A **PARTIAL** may not move to closed without runtime evidence (`/sys/fs/cgroup/.../memory.max readback, systemctl status` showing scope active, `kill -9 survivor` proof — never just script exit code).
- A **BLOCKED** reclassifies to **OPEN** when the blocking dependency resolves; it must NOT skip directly to closed.
- An **INVESTIGATED** item must record the captured forensic trace (file path / command output / log timestamp) — never speculation ("likely" / "probably" / "appears to" — see Constitution §11.4.6).

Categories:

- **A** — Tooling / harness gaps
 - **B** — Anti-bluff completeness across the existing test surface
 - **C** — Per-session containerization features pending evidence
 - **D** — Host-capability + topology dispatch gaps
 - **E** — Documentation / Continuation drift
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A. Tooling / harness gaps

A2 RUNALL-NATIVE-RESOLVE-001 — standalone run_all.sh mis-resolves the binary on native macOS

Status: Fixed (→ Fixed.md) **Type:** Task

CLOSED v1.0.27. scripts/tests/run_all.sh hardcodes TMUX_BIN=tmux/build/bin/tmux (the build_containerized.sh output path) and is the containerized-build validator. On native macOS the authoritative validator is setup.sh (builds + verifies against tmux/build-darwin/). Invoked standalone on macOS with a stale tmux/build/ present (a prior Linux containerized build), run_all.sh resolved the wrong-arch binary → Exec format error mass-FAIL (observed 2026-06-16; NOT a product defect — setup.sh run_all 55/0/6 + installed-binary smoke GREEN the same session; removing the stale tmux/build/ then gave not executable because run_all expects that path). **Fix direction:** make run_all.sh OS-aware (prefer tmux/build-darwin/ on Darwin) OR document that native-macOS validation is setup.sh-only and run_all.sh is the containerized path. Captured-evidence requirement: a clean native-macOS run_all GREEN after the fix.

A50 GO-TOOLCHAIN-OBTAIN-001 — obtain Go toolchain locally for the tmx-state + workable-items Go build — OPEN

TMX-ID: TMX-057 **Status:** OPEN **Type:** Task **Severity:** MEDIUM

On hosts lacking a system Go toolchain (e.g. amber), the scripts/tmx-state/ and cmd/workable-items/ Go binaries cannot be (re)built — blocking per-session cwd persistence AND the §11.4.93 workable-items SSoT tooling. **Fix direction:** extend the existing §11.4.77 local-deps mechanism (obtain_local_deps.sh, which already sha256-obtains libevent 2.1.12 + ncurses 6.5 into git-ignored .local-deps/) to resolve-or-obtain a pinned Go toolchain into .local-deps/, consumed by the Go build steps via GOROOT/PATH. **Acceptance:** on a host with no system go, a clean go build ./cmd/workable-items succeeds against the obtained local toolchain (exit 0; control without it → go: command not found).

A51 NATIVE-BUILD-CC-LINK-MSG-001 — native-build fallback surfaced a cryptic compiler error instead of an honest, actionable message — INVESTIGATED

TMX-ID: TMX-061 **Status:** Ready for testing **Type:** Bug **Severity:** HIGH

On a base ALT host where gcc is present but cannot link (glibc-devel missing), the setup.sh native-build fallback surfaced the raw cryptic autotools error "C compiler cannot create executables" instead of an honest, actionable message — a \$11.4.138 operator-escape the green suite missed. FIXED this session: setup.sh gains cc_can_link + _native_build_preflight emitting an honest diagnostic and auto-installing the missing toolchain when running as root, guarded by scripts/tests/70 (RED→GREEN) with a bluff-audit at docs/research/native_fallback_cc_link_bluff_audit/. Acceptance: RED_MODE=1 reproduces the cryptic failure on the broken artifact, RED_MODE=0 is GREEN, and the paired meta-mutation is CAUGHT — verified post-commit on a clean tree (test 70 currently PASS=16/0/0; pending commit + post-commit meta-validation).

A52 NO-SUDO-NO-INTERACTION-001 — no sudo/su execution and no user-interaction in any automation script or test — INVESTIGATED

TMX-ID: TMX-062 **Status:** Ready for testing **Type:** Task **Severity:** HIGH

Operator mandate 2026-06-29: no automation script or test may execute sudo/su or require human interaction. DONE this session — removed the single sudo execution, made install root-only, and reworded advice so setup.sh + install_deps.sh + install.sh contain 0 sudo/su tokens; added the verify.sh gate CM-NO-SUDO-NO-INTERACTION with paired meta-mutation M-CM-NO-SUDO-NO-INTERACTION (CAUGHT) and test 70 case C10. Acceptance: 0 sudo/su tokens AND 0 human-wait points across the in-scope files, and the gate FAILs when a sudo token is injected — pending commit + post-commit meta-validation on a clean tree.

A53 ROOT-FREE-LOCAL-BUILD-001 — fully-autonomous root-free local build (obtain all build deps locally; no root/sudo/interaction) — PARTIAL

TMX-ID: TMX-063 **Status:** In progress **Type:** Feature **Severity:** HIGH

Operator mandate 2026-06-29: bash scripts/setup.sh must build AND install tmux with NO root, NO sudo, and NO interaction by obtaining every mandatory build dependency locally. Research-backed plan (docs/research/root_free_c_toolchain_20260629/): obtain a Zig zig cc C toolchain into .local-deps (kind=toolchain, like the Go-toolchain obtain) and build tmux from the 3.6a release tarball to avoid autotools/system-cc requirements. Status In progress — research complete and a proof-of-concept live-build spike is running; implementation is pending the PoC verdict. Acceptance: a live root-free build on a real host with the system

toolchain neutered produces a tmux 3.6a binary that runs, with captured evidence and N=3 deterministic reproduction — no bluff.

A54 NO-SUDO-PROJECTWIDE-FOLLOWUP-001 — convert print-only sudo/setcap hints outside the install path and extend the no-sudo gate project-wide — OPEN

TMX-ID: TMX-064 **Status:** OPEN **Type:** Task **Severity:** MEDIUM

Follow-up to TMX-062: convert the remaining print-only sudo/setcap hints OUTSIDE the install path (scripts/build_oom_set.sh, scripts/test_vm.sh, scripts/tests/08_oom_score_adj.sh, and the scripts/oom_set.c comment) to "(as root)" phrasing, and extend the no-sudo gate project-wide so it detects sudo/su EXECUTION rather than mere mention. Status Queued. Acceptance: project-wide 0 sudo/su execution paths, and the gate is scoped to flag execution only — no false positives on legitimate "(as root)" documentation strings.

B. Anti-bluff completeness across the existing test surface

(Prior B-items closed: B3 P5-M20/P5-M21 escapes CLOSED in v1.0.16 [tests 49/50 + meta-test retarget], state-verified 2026-05-29 with MUTATIONS CAUGHT 45 / ESCAPED 0, migrated to Fixed.md §B3 as TMX-054; B1 CHAL-COVER-001, B2 TEST-AUDIT-001 also in Fixed.md. New open work below.)

B50 TEST-COVERAGE-G1-G5-001 — close test-coverage gaps G1-G5 for the v1.0.30 cross-platform install hardening — OPEN

TMX-ID: TMX-059 **Status:** OPEN **Type:** Task **Severity:** MEDIUM

v1.0.30 added native-build fallback (setup.sh), build_native.sh local-deps wiring (-I/-L + PKG_CONFIG_PATH), obtain_local_deps.sh libevent/ncurses obtain, and the escaped-colon session-color fix; the CHANGELOG tracks "test-coverage gaps G1-G5" but they are NOT yet enumerated distinctly nor each covered by an anti-bluff test + paired §1.1 mutation. **Fix direction:** first enumerate G1-G5 precisely, then add four-layer coverage (§11.4.4(b)) for each. **Acceptance:** each of G1-G5 has a named runtime test with captured evidence PLUS a paired meta-test mutation that FAILs when its guard is stripped.

B51 DB-FIXEDMD-SSOT-DRIFT-001 — reconcile DB↔Fixed.md SSoT drift (Fixed.md A46-A49 absent from the items table) — INVESTIGATED

TMX-ID: TMX-060 **Status:** Ready for testing **Type:** Task
Severity: MEDIUM

The §11.4.93 workable-items DB is missing four RESOLVED Fixed.md entries — A46 (libtinfo version warning), A47 (test-harness timing races), A48 (distribution orchestrator binary), A49 (test 17 scrollbar load flake). A full md-to-db over Fixed.md reports inserted=4 updated=1 allocated=4, proving the items table never captured them (the CHANGELOG names this "DB↔Fixed.md SSoT drift" as a tracked follow-up). **Fix direction:** reconcile carefully — confirm none are reworded duplicates of an existing id before allocating, identify the updated=1 item, then sync so validate + diff stay clean. **Acceptance:** md-to-db over the full corpus reports inserted=0 updated=0 (idempotent) and validate is 0-findings. **Reconcile (2026-06-29, this session):** ran sync md-to-db over the full corpus — the four entries ingested as TMX-067 (A49 → Completed), TMX-068 (A48 → Implemented), TMX-069 (A46 → Fixed) and TMX-070 (A47 → Fixed); the updated=1 item was TMX-045 ("Per-host-topology dispatch probe") whose stored raw_body was stale (2140→7648 chars) and was refreshed from the current Fixed.md — no other field changed, so it is not a reworded duplicate. The run also re-captured the stale document_sources[Fixed] so a future db-to-md no longer regresses Fixed.md. A second md-to-db reports inserted=0 updated=0 (idempotent) and validate is 0-findings — pending conductor verification + close.

B52 WI-PARSER-GREEDY-BIND-001 — workable-items md-parser absorbs a following no-period block's TMX-ID → UNIQUE-constraint failure — INVESTIGATED

TMX-ID: TMX-065 **Status:** Ready for testing **Type:** Bug **Severity:** HIGH

cmd/workable-items/parser.go greedily extended a period-style heading's (### A54. ..., which matches headingRE) 24-line structured-metadata window across any FOLLOWING no-period ### block, mis-binding that block's **TMX-ID:** / **Type:** / **Status:** to the period item — so sync md-to-db aborted with UNIQUE constraint failed: items.atm_id (forced a §9.2 DB restore this session). FIXED this session: the parser now closes the structured-metadata window at the first subsequent Markdown heading of ANY level (anyHeadingRE), so a following block's metadata can never be absorbed by a preceding period item. Guarded by cmd/workable-items/parser_greedybind_test.go

RED→GREEN (parser-level absorption test + end-to-end UNIQUE-collision regression). **Acceptance:** a period heading never absorbs a following block's TMX-ID/Type and md-to-db raises no UNIQUE error — go test ./cmd/workable-items GREEN; pending conductor verification + close.

B53 WI-SET-STATUS-COMMAND-001 — workable-items had no command to set a non-terminal status (manual sqlite3 UPDATE required) — INVESTIGATED

TMX-ID: TMX-066 **Status:** Ready for testing **Type:** Bug **Severity:** MEDIUM

workable-items add only ever creates a Queued item and close only ever assigns a TERMINAL closure status, so moving an item to a non-terminal §11.4.15/§11.4.21 status (In progress / Ready for testing / In testing / Reopened / Operator-blocked) required a hand-written sqlite3 UPDATE that bypassed the §11.4.34 audit trail and the §11.4.93 SSoT discipline. FIXED this session: new workable-items set-status TMX-NNN --status <non-terminal> [--by AI|User] [--reason ...] writes the status + last_modified + an audited Updated item_history row, rejects terminal statuses (pointing to close) and unknown statuses (§11.4.6 — no silent default). Guarded by cmd/workable-items/set_status_test.go (6 cases). **Acceptance:** the command sets a non-terminal status with an audit row and rejects terminal/unknown values — go test ./cmd/workable-items GREEN; pending conductor verification + close.

C. Per-session containerization features pending evidence

(none open at this time — C1 TMX-T5, C2 TMX-T7, C3 TMX-T8 landed in Fixed.md.)

D. Host-capability + topology dispatch gaps

D2 TMPDIR-HARDCODE-001 — tests hardcoding / tmp false-FAIL under host disk-pressure

Status: Fixed (→ Fixed.md) **Type:** Task

CLOSED v1.0.27. Several tests create scratch under a hardcoded /tmp (e.g. 27_state_persistence.sh target tmp-test-18-target-*). When the host root volume is full (observed 2026-06-16 on

macOS, / at <200 MiB during the operator's away-window), the cd/mkdir into /tmp fails → pane_current_path='' false-FAIL instead of an honest §11.4.3 SKIP-with-reason. The tests pass normally + standalone (27 3/3, 38 3/3, 43 15/0 re-run the same session) — the failure is purely the abnormal host-disk condition (a §11.4.1 FAIL-bluff class: environment, not product). **Fix direction:** route test scratch through \$TMPDIR (operator now sets /Volumes/T7/tmp via ~/.zshrc + ~/.bashrc) OR guard each test on /tmp writability and SKIP-with-reason (§11.4.3/§11.4.50). Captured-evidence requirement: an induced-disk-full run shows SKIP-with-reason, not FAIL.

E. Documentation / Continuation drift

(CONTINUATION.md §3 entries that resolve land in Fixed.md per Constitution §5 / §12.10. New open work below.)

E50 INSTALL-DOC-PODMAN-HTTPS-001 — document the rootless-Podman subuid fix and the curl-installer HTTPS-rewrite edge — OPEN

TMX-ID: TMX-058 **Status:** OPEN **Type:** Task **Severity:** LOW

v1.0.30 shipped a native-build fallback when rootless Podman exhausts /etc/subuid + /etc/subgid (lchown ... invalid argument) and a curl one-liner installer, but the operator-facing repair guidance lives only in the CHANGELOG. **Fix direction:** document the usermod --add-subuids/--add-subgids + podman system migrate repair recipe AND the install HTTPS-rewrite edge in docs/guides + docs/scripts so an end user hitting either can self-recover.

Acceptance: a docs page (synced HTML/PDF per §11.4.65) reproduces the repair steps, verified against the v1.0.30 setup.sh fallback path.

F. Runtime crash — operator-gated reproduction

(none open — F1 tmx session named "HelixCode" crashes the whole terminal RESOLVED 2026-06-13, operator-confirmed, migrated to Fixed.md A45 with reproduced root cause [stale wrapper TMUX_BIN pointing at a non-existent prior-checkout path → exec of missing binary → login shell dies → terminal closes] + 4-layer regression guard [test 60 + verify gate CM-TMX-WRAPPER-TMUXBIN-VALID + meta M-WRAPPER-TMUXBIN]).

Last reviewed: 2026-06-19 (v1.0.27 WIP burn-down. §11.4.138 operator-escape A51 [name:color prompt rejection] FIXED — shell-init + SSH-dispatcher char-set now allows : and #, max-length 64→80; test 65 operator-escape guard PASS=6/6. M24/A2/D2 in parallel streams per §11.4.103.)

M24-ESCAPE-001 — meta-test M24 (hostname 4-surface color) escapes: test 26 misses a 3-set-line removal

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Minor (test-coverage gap, no user-facing break — closed v1.0.27: count=1 removed from M24 regex → strips from BOTH `_apply_color` and `_apply_host_color`, test 26 now FAILs on the mutation → CAUGHT) **Closure:** meta-test 37 CAUGHT / 0 ESCAPED (was 34 CAUGHT / 3 ESCAPED pre-fix)

What: The §1.1 paired-mutation M24 in `scripts/tests/meta_test_false_positive_proof.sh` removes three of the four `tmux set -g ...` lines in `_apply_host_color()` and expects test 26 to FAIL. The harness reports `MUTATION ESCAPED` — test 26 does not FAIL, because it asserts only a subset of the four surfaces, so removing the un-asserted set-lines leaves the test green.

Evidence: `bash scripts/tests/meta_test_false_positive_proof.sh 2>&1 | grep M24 → FAIL: M24: MUTATION ESCAPED` — test 26 did not FAIL with the three set-lines removed. Confirmed pre-existing (M24 added in commit f151d13, v1.0.9 — before the per-session-color feature). Surfaced 2026-06-19 during the per-session-color M25/M26 verification run.

Fix direction: strengthen test 26 to assert ALL FOUR surfaces (`status-style`, `pane-active-border-style`, `clock-mode-colour`, `window-status-current-style`) so the M24 mutation (removing any 3) reliably FAILs it — mirroring the all-4-surfaces assertion already proven in test 63 T3 for the per-session path. (This also closes the symmetry gap: the per-session path has a 4-surface guard; the `hostname` path should too.)

Out of scope: the per-session-color feature (TMX-051). Tracked separately so the color release is not blocked by an unrelated pre-existing test-gap.